Production Specification

Trend Identifier v2.4

Full Reference Specification

Document Type	Technical specification
Version	v2.4
Audience	Engineering, research, platform, risk, data, technical product owners, and internal reviewers.
Public Output	UP | FLAT | DOWN
Purpose	Define a production-grade multi-timeframe Trend Identifier that classifies an instrument as UP, FLAT, or DOWN.
Scope	This document specifies the full reference design, including objective and definitions, data requirements and eligibility gates, multi-timeframe architecture, exact feature definitions, formulas, thresholds, rule logic, hysteresis, futures handling, validation framework, formal JSON Schema, canonical reason codes, function contracts, deterministic missing-value handling, runtime guarantees, and appendices required for implementation and audit.
Implementation Posture	This is the standalone normative reference specification for production implementation, validation, audit, and technical review. It is intended to be self-contained, audit-friendly, and directly usable by its intended audience.
Normative Status	This is a standalone full reference document intended to be read on its own. It does not rely on prior versions for interpretation.

1. Executive Summary

The Trend Identifier is a hierarchical, rule-based regime classifier for liquid stocks and liquid index or futures instruments. It classifies the instrument at each evaluation timestamp into exactly one public label: UP, FLAT, or DOWN. The classifier is intended for production decision support and auditability rather than next-bar forecasting or discretionary chart commentary.

The engine combines weekly, daily, and hourly candles. Weekly defines structural direction, daily confirms or weakens that structure, and hourly detects near-term alignment or deterioration. The design is score-gated, persistence-filtered, explainable, and explicitly resistant to noise, event shocks, and low-quality inputs.

This document is the full standalone reference specification. It contains the complete objective, definitions, formulas, thresholds, rule logic, hysteresis, futures handling, pseudocode, test scenarios, edge-case handling, formal JSON Schema, canonical reason-code vocabulary, strict function contracts, deterministic missing-value handling, and execution/runtime guarantees required for implementation and audit.

2. Locked Definitions

- **UP:** Evidence supports a stable upward regime.
- **DOWN:** Evidence supports a stable downward regime.
- **FLAT:** Directional evidence is weak, conflicted, unreliable, or insufficiently persistent.
- **UNCLASSIFIABLE:** Internal state used when minimum history, liquidity, or data-quality requirements fail.
- **Confidence:** A heuristic trust score in the issued public label. It is not a calibrated probability.
- **Regime strength:** Directional-dominance magnitude irrespective of whether a veto later downgrades the public label to FLAT.
- **Transition clock:** Daily evaluation close is the only public-label commit point. Hourly evaluations may update diagnostics and confidence intraday but do not commit public-label transitions.

3. Exact Data Requirements and Eligibility Gate

The engine shall run only on instruments that pass all hard checks. If any hard check fails, set internal_state = UNCLASSIFIABLE. In the reference implementation, the engine still emits a complete public payload: label MUST be FLAT, confidence MUST be <= 0.20, regime_strength MUST equal 0.0, aggregate_score MUST be null, and per-timeframe score components MUST be null.

All timestamps must be normalized to the instrument exchange calendar. Missing bars may be calendar-adjusted, but actual trading gaps must never be forward-filled.

Check	Default rule	Type	Failure action
Weekly history	at least 104 completed weekly bars	Hard	UNCLASSIFIABLE
Daily history	at least 252 completed daily bars	Hard	UNCLASSIFIABLE
Hourly history	at least 480 completed hourly bars	Hard	UNCLASSIFIABLE
Missing bars	<= 3% in last 60 daily bars and <= 5% in last 240 hourly bars after calendar adjustment	Hard	UNCLASSIFIABLE
Liquidity: equities	median 20-day dollar volume >= USD 5 million equivalent	Hard	UNCLASSIFIABLE
Liquidity: index futures	median 20-day notional volume >= instrument-family threshold and open interest above venue minimum	Hard	UNCLASSIFIABLE
Bad candles	reject if High < max(Open,Close), or Low > min(Open,Close), or any non-positive price appears	Hard	UNCLASSIFIABLE
Staleness	latest completed bar timestamp within one expected bar interval of evaluation time	Hard	UNCLASSIFIABLE

4. Multi-Timeframe Design

- **Weekly:** structural anchor; 50% final aggregation weight.
- **Daily:** confirmation layer; 35% final aggregation weight.
- **Hourly:** refinement and deterioration layer; 15% final aggregation weight.

Timeframe precedence is weekly > daily > hourly. A lower timeframe may reduce confidence or force FLAT in border cases, but it does not reverse a strong higher-timeframe regime on its own. Direct contradiction between weekly and daily is treated as a major conflict and usually resolves to FLAT.

5. Exact Feature Definitions

Per-timeframe lookbacks are fixed defaults: weekly = 8, 13, 26, 52 weeks; daily = 10, 20, 60, 120 days; hourly = 12, 24, 48, 120 hours. ATR uses Wilder ATR(14) on the same timeframe. Notation: close_t, open_t, high_t, low_t denote latest completed-bar values on timeframe T. EMA_n is the exponential moving average with span n. HHV_L and LLV_L are the highest high and lowest low over the prior L completed bars, excluding the current bar. Unless otherwise stated, all ratios are computed on the adjusted feature-computation series for the relevant instrument type.

5.1 Exact realized-volatility definition

Return definition: r_i = ln(close_i / close_{i-1}).

realized_vol_L = sqrt(252) * stdev(r_{t-L+1}, ..., r_t) for weekly and daily; realized_vol_L = sqrt(252 * session_bars_per_day) * stdev(r_{t-L+1}, ..., r_t) for hourly. For US-style 6.5-hour cash sessions, session_bars_per_day = 6; for 24-hour futures, use the house-standard completed bars per exchange day and keep it fixed by instrument family.

If annualization is not desired in implementation, the engine may use raw standard deviation instead, but the choice must be fixed consistently across VOLX numerator and denominator. The reference implementation uses annualized realized volatility.

5.2 Direction features

NR_T = (close_t - close_{t-L_ret}) / ATR14_t, where L_ret = 13 weeks, 60 days, or 48 hours.

BS_T = (EMA_fast_t - EMA_fast_{t-L_slope}) / (L_slope * ATR14_t), where EMA_fast = 13, 20, or 24 and L_slope = 4 weeks, 10 days, or 12 hours.

SEP_T = (EMA_fast_t - EMA_slow_t) / ATR14_t, where EMA_slow = 26, 60, or 120.

BOD_up_T = (close_t - HHV_{L_break}) / ATR14_t and BOD_dn_T = (LLV_{L_break} - close_t) / ATR14_t, where L_break = 26 weeks, 60 days, or 48 hours.

SW_T = +1 if the last two confirmed pivots are higher-high and higher-low; SW_T = -1 if they are lower-high and lower-low; else SW_T = 0.

5.3 Exact tie rules for pivots and breakout acceptance

Pivot confirmation: hourly and weekly use a 2-2 fractal; daily uses a 3-3 fractal. A pivot high at index j requires high_j strictly greater than all highs in the lookback window on both sides. A pivot low requires low_j strictly lower than all lows in the lookback window on both sides.

Ties are not pivots. If any candidate pivot high is equal to the maximum high of its left-right confirmation window, discard that candidate. Likewise, if any candidate pivot low is equal to the minimum low of its confirmation window, discard that candidate.

Higher-high and lower-high comparisons also use strict inequality. Equal highs or equal lows do not count as directional swing confirmation; they map to neutral for SW_T unless later bars create a strict breakout.

breakout_sign_T = +1 only if both conditions hold: (close_t >= HHV_{L_break} + 0.25 * ATR14_t) AND (ACC_breakout_up_T >= 0.60). breakout_sign_T = -1 only if (close_t <= LLV_{L_break} - 0.25 * ATR14_t) AND (ACC_breakout_dn_T >= 0.60). Otherwise breakout_sign_T = 0.

ACC_breakout_up_T is the fraction of the next N_acc closes after the first qualifying breakout that remain >= HHV_{L_break}. ACC_breakout_dn_T is defined symmetrically using <= LLV_{L_break}. If fewer than N_acc completed closes exist because the breakout is too recent, compute the fraction on available closes and mark acceptance_partial = true in diagnostics; do not upgrade from FLAT to directional on partial acceptance alone.

5.4 Quality and noise features

EFF_T = abs(close_t - close_{t-L_eff}) / sum_{i=t-L_eff+1}^{t} abs(close_i - close_{i-1}), where L_eff = 13 weeks, 40 days, or 36 hours.

ACC_T = fraction of last N_acc closes that are on the correct side of EMA_slow_t if no valid breakout exists; otherwise use the breakout acceptance fraction against the relevant breakout level. N_acc = 6 weeks, 15 days, or 12 hours.

OVL_T = 1 - abs(median(close over N_ovl) - close_t) / max((max(high over N_ovl) - min(low over N_ovl)), 0.5 * ATR14_t), clipped to [0,1], where N_ovl = 8 weeks, 20 days, or 24 hours.

GAP_T = abs(open_t - close_{t-1}) / ATR14_t. VOLX_T = realized_vol_10 / realized_vol_60 for daily and hourly, and realized_vol_4 / realized_vol_26 for weekly. ANOM_T = 1 if true_range_t > 4 * median(true_range over prior 20 bars), else 0. LIQ_T = 1 if current 20-bar median dollar volume is below threshold, else 0.

6. Exact Score Construction

All intermediate component scores are clipped to [-1, +1] before weighting. zclip(x) = min(max(x, -1), +1).

direction_score[T] = 0.35 * zclip(NR_T / K_NR) + 0.25 * zclip(BS_T / K_BS) + 0.20 * zclip(SEP_T / K_SEP) + 0.10 * SW_T + 0.10 * breakout_sign_T

quality_score[T] = 0.50 * zclip((EFF_T - EFF_min_T) / (1 - EFF_min_T)) + 0.30 * zclip((ACC_T - ACC_min_T) / (1 - ACC_min_T)) - 0.20 * OVL_T

noise_score[T] = 0.35 * zclip(GAP_T / GAP_veto_T) + 0.35 * zclip((VOLX_T - 1) / (VOLX_veto_T - 1)) + 0.20 * ANOM_T + 0.10 * LIQ_T

timeframe_score[T] = 100 * (0.55 * direction_score[T] + 0.30 * quality_score[T] - 0.15 * noise_score[T])

Timeframe	K_NR	K_BS	K_SEP	EFF_min	ACC_min
Weekly	6.0	0.12	1.50	0.38	0.67

Daily	4.0	0.10	1.20	0.34	0.60
Hourly	3.0	0.08	1.00	0.30	0.58
Timeframe	**UP_THRESHOLD**	**DOWN_THRESHOLD**	**QUALITY_MIN**	**GAP_veto**	**VOLX_veto**
Weekly	+22	-22	0.15	2.0 ATR	1.80
Daily	+18	-18	0.10	2.0 ATR	1.75
Hourly	+15	-15	0.05	1.8 ATR	1.60

7. Parenthesized Rule Logic

IF ((timeframe_score[T] >= UP_THRESHOLD[T]) AND (quality_score[T] >= QUALITY_MIN[T])) THEN provisional_label[T] = UP.

ELSE IF ((timeframe_score[T] <= DOWN_THRESHOLD[T]) AND (quality_score[T] >= QUALITY_MIN[T])) THEN provisional_label[T] = DOWN.

ELSE provisional_label[T] = FLAT.

major_conflict = ((weekly_label == UP) AND (daily_label == DOWN)) OR ((weekly_label == DOWN) AND (daily_label == UP))

score_conflict = ((sign(weekly_score) != sign(daily_score)) AND (abs(weekly_score) >= 12) AND (abs(daily_score) >= 12))

quality_conflict = ((daily_label == weekly_label) AND (daily_quality_score < 0) AND (((hourly_score <= -18) AND (weekly_label == UP)) OR ((hourly_score >= +18) AND (weekly_label == DOWN))))

aggregate_score = (0.50 * weekly_score) + (0.35 * daily_score) + (0.15 * hourly_score)

Rule A: IF ((weekly_label == FLAT) AND NOT (((weekly_score >= +18) AND (daily_score >= +10)) OR ((weekly_score <= -18) AND (daily_score <= -10)))) THEN final_candidate = FLAT.

Rule B: ELSE IF ((major_conflict == true) OR (score_conflict == true)) THEN final_candidate = FLAT.

Rule C: ELSE IF ((any_veto_flag == true) AND (abs(aggregate_score) < 35)) THEN final_candidate = FLAT.

Rule D: ELSE IF ((weekly_label == UP) AND (daily_label == UP) AND NOT ((hourly_label == DOWN) AND (hourly_score <= -20))) THEN final_candidate = UP.

Rule E: ELSE IF ((weekly_label == DOWN) AND (daily_label == DOWN) AND NOT ((hourly_label == UP) AND (hourly_score >= +20))) THEN final_candidate = DOWN.

Rule F: ELSE IF (aggregate_score >= +25) THEN final_candidate = UP.

Rule G: ELSE IF (aggregate_score <= -25) THEN final_candidate = DOWN.

Rule H: ELSE final_candidate = FLAT.

shock_veto = (((GAP_daily >= 2.0) OR (GAP_weekly >= 2.0)) AND (ACC_daily < 0.60))

volatility_veto = ((VOLX_daily >= 1.75) AND (EFF_daily < 0.30))

anomaly_veto = ((ANOM_daily == 1) AND (ANOM_hourly == 1))

liquidity_veto = ((LIQ_daily == 1) OR (LIQ_hourly == 1))

any_veto_flag = shock_veto OR volatility_veto OR anomaly_veto OR liquidity_veto OR roll_veto

8. Exact Hysteresis and Transition Rules

IF (final_candidate == previous_final_label) THEN keep label and set transition_state = stable.

IF ((previous_final_label == FLAT) AND (final_candidate == UP)) THEN require (((up_candidate_count_consecutive >= 2)) OR ((aggregate_score >= +40) AND (weekly_score >= +25))).

IF ((previous_final_label == FLAT) AND (final_candidate == DOWN)) THEN require (((down_candidate_count_consecutive >= 2)) OR ((aggregate_score <= -40) AND (weekly_score <= -25))).

IF ((previous_final_label == UP) AND (final_candidate == DOWN)) THEN force one intermediate FLAT state UNLESS (((aggregate_score <= -50) AND (weekly_score <= -25) AND (daily_score <= -25)) for two consecutive daily evaluations).

IF ((previous_final_label == DOWN) AND (final_candidate == UP)) THEN symmetric rule applies.

IF (((previous_final_label == UP) OR (previous_final_label == DOWN)) AND (final_candidate == FLAT)) THEN switch immediately IF ((abs(aggregate_score) < 15) OR (any_veto_flag == true)); ELSE require one confirming daily evaluation.

Hourly-only deterioration may lower confidence immediately but cannot trigger a structural reversal. Daily close is the only transition commit point.

9. Futures and Continuous-Series Handling

Use a continuous back-adjusted series for feature computation and a front-contract execution series for operational reporting.

Roll from contract n to contract n+1 when ((avg_volume_next_3d > avg_volume_front_3d) AND (open_interest_next >= 0.80 * open_interest_front)).

Apply additive back-adjustment for index futures. Ratio-adjustment is allowed only if fixed consistently for the instrument family and documented in metadata. All ATR, EMA, slope, breakout, and volatility features must use the adjusted feature-computation series. Liquidity checks must use the live front contract.

During the two sessions around a roll, set roll_flag = 1 and cap hourly aggregation weight at 0.05 instead of 0.15. If adjusted-series roll gap exceeds 1.5 ATR, set data_quality_warning = true and suppress directional upgrades for one full daily cycle.

10. Formula Appendix

true_range_t = max(high_t - low_t, abs(high_t - close_{t-1}), abs(low_t - close_{t-1}))

ATR14_t uses Wilder smoothing: ATR_t = ((ATR_{t-1} * 13) + true_range_t) / 14 after the initial simple-average seed.

EMA_n uses alpha = 2 / (n + 1).

sign(x) = +1 if x > 0, -1 if x < 0, 0 if x = 0.

clip01(x) = min(max(x, 0), 1).

confidence = clip01((0.35 * alignment) + (0.25 * abs(aggregate_score) / 50) + (0.20 * max(weekly_quality, 0)) + (0.10 * max(daily_quality, 0)) + (0.10 * (1 - veto_count / 4)))

alignment = 1.0 if weekly_label == daily_label == hourly_label in sign-consistent form; alignment = 0.5 if weekly and daily agree but hourly is FLAT or mildly contradictory; alignment = 0.0 if major_conflict is true.

regime_strength = tanh(abs(aggregate_score) / 40).

11. Formal JSON Schema and Public Payload Policy

Reference policy: the engine always emits a public payload; there is no suppressed-label mode in the reference implementation. If internal_state = CLASSIFIABLE, label is the model result: UP, FLAT, or DOWN. If internal_state = UNCLASSIFIABLE, label MUST be FLAT, confidence MUST be in [0.00, 0.20], regime_strength MUST equal 0.0, aggregate_score MUST be null, and per-timeframe score components MUST be null. Downstream consumers MUST NOT remap the public label when internal_state = UNCLASSIFIABLE.

The schema below is normative. It is expressed as JSON Schema draft 2020-12.

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      "TRANSITION_PENDING_DOWNGRADE",
      "TRANSITION_FORCED_FLAT",
      "RUNTIME_SCHEMA_VALIDATION_FAILED",
```

```
            "RUNTIME_STATE_PERSIST_FAILED",
            "RUNTIME_LOG_WRITE_FAILED",
            "RUNTIME_PREVIOUS_STATE_LOCK_FAILED"
          ]
        },
        "uniqueItems": true
      }
    }
  },
  "error": {
    "type": "object",
    "additionalProperties": false,
    "required": [
      "code",
      "message"
    ],
    "properties": {
      "code": {
        "type": "string",
        "minLength": 1
      },
      "message": {
        "type": "string",
        "minLength": 1
      }
    }
  }
 }
}
```


11.1 Canonical reason-code enumeration

Code	Meaning
GATE_HISTORY_WEEKLY	Fewer than 104 completed weekly bars.
GATE_HISTORY_DAILY	Fewer than 252 completed daily bars.

GATE_HISTORY_HOURLY	Fewer than 480 completed hourly bars.
GATE_MISSING_BARS	Missing-bar tolerance breached after calendar adjustment.
GATE_LIQUIDITY	Liquidity threshold not met.
GATE_BAD_CANDLE	Invalid candle structure or non-positive prices.
GATE_STALE_DATA	Latest completed bar is stale.
NUMERIC_INVALID	Any formula generated NaN or Inf.
MISSING_INTERMEDIATE_LOAD_BEARING	Load-bearing feature could not be computed.
REALIZED_VOL_INSUFFICIENT	Not enough completed returns for realized-volatility window.
ACCEPTANCE_PARTIAL	Breakout acceptance computed on partial future window only.
DIAGNOSTIC_MISSING	Non-load-bearing diagnostic missing only.
CONFLICT_MAJOR	Weekly and daily provisional labels are opposite.
CONFLICT_SCORE	Weekly and daily signed scores conflict materially.
CONFLICT_QUALITY	Daily quality failed with adverse hourly contradiction.
VETO_SHOCK	Shock veto active.
VETO_VOLATILITY	Volatility veto active.
VETO_ANOMALY	Anomaly veto active.
VETO_LIQUIDITY	Liquidity veto active.
VETO_ROLL	Roll-period veto or roll-upgrade suppression active.
TRANSITION_PENDING_UPGRADE	Candidate upgrade exists but persistence not satisfied.
TRANSITION_PENDING_DOWNGRADE	Candidate downgrade exists but persistence not satisfied.
TRANSITION_FORCED_FLAT	Immediate or required transition to FLAT.
RUNTIME_SCHEMA_VALIDATION_FAILED	Payload failed schema validation; publication blocked.
RUNTIME_STATE_PERSIST_FAILED	State persistence failed; payload not committed.
RUNTIME_LOG_WRITE_FAILED	Decision log write failed.
RUNTIME_PREVIOUS_STATE_LOCK_FAILED	Could not acquire or verify prior-state lock/version.

12. Strict Function Contracts

Function	Inputs	Outputs	Contract
load_and_align_data	instrument_id, asof_time, calendar, raw_bars	aligned weekly/daily/hourly bars	Must return completed bars only; no forward-fill across real trading gaps; must tag adjusted missing bars.
passes_eligibility_gate	aligned bars, liquidity metadata	gate_result	Must return deterministic pass/fail plus reason_codes and errors.
build_feature_series	aligned bars, instrument metadata	feature series bundle	Must return adjusted computation series and live

			front-contract liquidity series for futures.
compute_features	feature series bundle	feature object	Must compute all features or explicit nulls under Section 13 rules; no silent coercion.
compute_timeframe_scores	feature object, constants	score object	Must apply locked formulas exactly and clip intermediate values exactly.
assign_provisional_labels	score object, thresholds	provisional labels	Must follow parenthesized rules in Section 7 exactly.
compute_veto_flags	feature object, provisional labels, scores	veto object	Must compute all booleans explicitly; no inferred defaults.
compute_conflicts	provisional labels, scores	conflict object	Must emit major, score, quality conflicts.
apply_aggregation_rules	provisional labels, scores, vetoes, conflicts	final candidate	Must evaluate rules in documented order only.
apply_hysteresis	candidate, previous state, scores, vetoes	final label, transition state	Must commit transitions only on daily close.
package_diagnostics	all intermediate objects	diagnostics object	Must be lossless for all load-bearing fields used by audit and debugging.
public_output	final label, confidence, regime_strength, diagnostics	JSON payload	Must conform to the formal JSON Schema exactly.

13. Deterministic NaN and Missing-Intermediate Handling

Condition	Required handling	Reason code / flag	Public effect
Hard-gate failure before feature computation	Set internal_state = UNCLASSIFIABLE; set aggregate_score and timeframe scores to null	error + reason_code	Public payload emitted with FLAT label under policy
Single feature undefined due to insufficient completed bars inside otherwise classifiable window	Treat as hard-gate failure if missing value is load-bearing for score construction	MISSING_INTERMEDIATE_LOAD_BEARING	UNCLASSIFIABLE
Breakout acceptance partial because breakout is too recent	Compute available fraction; set acceptance_partial = true; prohibit FLAT -> directional upgrade on partial acceptance alone	ACCEPTANCE_PARTIAL	May hold FLAT
Non-load-bearing diagnostic missing	Emit explicit null in diagnostics only	DIAGNOSTIC_MISSING	No forced label change
Any formula generates NaN or inf due to divide-by-zero or invalid input	Raise deterministic data-quality error; do not clip and do not coerce to zero	NUMERIC_INVALID	UNCLASSIFIABLE
Range denominator in OVL_T below 0.5 * ATR14_t	Use max(range, 0.5 * ATR14_t) exactly as specified	none	Continue
realized_vol window contains fewer than L completed returns	Fail feature computation for that timeframe	REALIZED_VOL_INSUFFICIENT	UNCLASSIFIABLE

14. Execution and Runtime Guarantees

- **Determinism:** Given identical completed bars, metadata, constants, and prior committed state, the engine must emit identical payloads.
- **Idempotency:** Re-running the same evaluation for the same instrument and asof_time must not create divergent state or duplicate committed transitions.
- **Publication order:** Schema validation must succeed before payload publication and before state commit.
- **State commit policy:** If persistence fails, the payload must not be marked committed; emit RUNTIME_STATE_PERSIST_FAILED.
- **Previous-state integrity:** Transition logic must use a verified prior-state version or lock; on failure emit RUNTIME_PREVIOUS_STATE_LOCK_FAILED and do not commit a transition.
- **Logging guarantee:** The engine must attempt machine-readable decision logging for every evaluation. If logging fails, emit RUNTIME_LOG_WRITE_FAILED.
- **Failure isolation:** A runtime failure for one instrument evaluation must not silently corrupt another instrument state.
- **No silent fallback:** The engine must not substitute zeros, stale scores, or stale prior payloads in place of failed computations.

15. Reference Pseudocode

```
function evaluate_trend(instrument, asof_time):
    data = load_and_align_data(instrument, asof_time)
    gate = passes_eligibility_gate(data)
    if gate.pass == false:

        internal_state = "UNCLASSIFIABLE"
        return emit_unclassifiable(gate, instrument, asof_time)

    internal_state = "CLASSIFIABLE"

    series = build_feature_series(data, instrument.metadata)
    feats = compute_features(series)
    assert_no_invalid_numeric(feats)

    scores = compute_timeframe_scores(feats, CONSTANTS)
    provisional = assign_provisional_labels(scores, CONSTANTS.thresholds)
    vetoes = compute_veto_flags(feats, provisional, scores)
    conflicts = compute_conflicts(provisional, scores)

    candidate = apply_aggregation_rules(
        provisional=provisional,
        scores=scores,
        vetoes=vetoes,
        conflicts=conflicts
    )

    final_label, transition_state = apply_hysteresis(
        candidate=candidate,
        previous_state=load_previous_state(instrument),
        scores=scores,
        vetoes=vetoes,
        asof_time=asof_time
    )

    confidence = compute_confidence(final_label, provisional, scores, vetoes)
    regime_strength = tanh(abs(scores.aggregate_score) / 40) if internal_state == "CLASSIFIABLE" else 0.0
    diagnostics = package_diagnostics(
```

```
        feats=feats,
        scores=scores,
        provisional=provisional,
        vetoes=vetoes,
        conflicts=conflicts,
        transition_state=transition_state
    )

    payload = public_output(...)
    validate_schema(payload)
    log_decision(instrument, asof_time, payload)
    persist_state(instrument, payload)
    return payload
```

16. Unit-Test Scenarios

ID	Scenario	Input condition	Expected result	Why it matters
UT-01	Eligibility fail	daily bars = 251	UNCLASSIFIABLE	hard gate
UT-02	Weekly/daily major conflict	weekly=UP, daily=DOWN	FLAT candidate	conflict handling
UT-03	Directional agreement	weekly=UP, daily=UP, hourly=FLAT, no veto, aggregate=+31	UP	core happy path
UT-04	Shock veto	GAP_daily=2.3, ACC_daily=0.45	FLAT unless prior live-regime exception applies	distortion control
UT-05	FLAT to UP fast-track	previous=FLAT, candidate=UP, aggregate=+42, weekly=+27	UP	hysteresis fast-track
UT-06	UP to DOWN direct reversal blocked	previous=UP, candidate=DOWN for one day only	FLAT intermediate	anti-whipsaw
UT-07	Pivot tie	equal highs inside pivot window	no pivot confirmation	tie rule
UT-08	Partial breakout acceptance	breakout too recent, only 4 of 12 closes available	no FLAT->UP upgrade on partial acceptance alone	acceptance strictness

17. Validation Framework with Exact Metrics

- Out-of-sample design: walk-forward. Use at least 5 chronological folds. Calibration is allowed only on prior folds.
- Turnover metric: annualized label changes per instrument. Report median and 90th percentile across the universe.
- Direct reversal metric: fraction of transitions that move UP -> DOWN or DOWN -> UP without an intermediate FLAT. Target should be near zero in baseline operation.
- Label persistence metric: median consecutive days spent in each label state.
- False directional break metric: among new UP labels, fraction where 20-day forward return is <= -1 ATR(20); among new DOWN labels, fraction where 20-day forward return is >= +1 ATR(20).

- FLAT miss metric: fraction of FLAT labels followed by absolute 20-day move >= 2 ATR(20) with efficiency >= 0.45.
- Chop capture metric: within externally defined range-bound windows, share of observations labeled FLAT.
- Shock robustness metric: within event windows where GAP_daily >= 1.5 ATR or VOLX_daily >= 1.5, measure change in false directional break rate versus normal periods.
- Sector diversity metric for equities: report all metrics by GICS-style sector or house sector mapping.
- Futures robustness metric: report metrics separately for equity index futures, rates, FX, commodities, and roll-adjacent periods.

No single metric is sufficient. A candidate calibration is acceptable only if it improves false directional breaks without causing excessive FLAT overuse or label churn.

18. Edge-Case Matrix

Edge case	Required behavior	Diagnostic flag	Public label effect
Roll-adjacent futures session	cap hourly weight to 0.05	roll_flag=1	reduce flip risk
Adjusted-series roll gap > 1.5 ATR	suppress directional upgrades for one daily cycle	data_quality_warning=true	hold or FLAT bias
Missing hourly bars within tolerance	calendar-adjust and continue	missing_bars_adjusted=true	no forced change
Actual trading gap forward-filled by vendor	reject feed	data_quality_error=true	UNCLASSIFIABLE
Equal high/low plateau	do not award pivot or breakout tie credit	pivot_tie=true	neutralize structure
Hourly contradiction only	reduce confidence; do not reverse structure	hourly_deterioration=true	keep higher-timeframe state unless other rules trip
Liquidity falls below threshold intraday	set liquidity veto	liquidity_veto=true	usually FLAT if aggregate weak

19. Rationale Appendix for Locked Constants

Constant group	Locked default	Rationale
Aggregation weights	Weekly 50%, daily 35%, hourly 15%	Keeps structural regime dominant while still allowing medium-horizon confirmation and intraday deterioration to matter.
EMA spans	13/26 weekly, 20/60 daily, 24/120 hourly	Pairs a responsive baseline with a slower anchor while keeping scales roughly proportional across timeframes.
Directional thresholds	+22, +18, +15 and symmetric negatives	Tighter threshold on faster frames would overclassify noise; looser threshold on slower frames would miss established regime.
Quality minima	0.15, 0.10, 0.05	Require at least modest path cleanliness while allowing trend signals in naturally noisier faster frames.
Fast-track hysteresis	+40 and -40 aggregate with weekly +/-25	Allows decisive transitions when evidence is strong without letting one ordinary day flip regime.

Direct reversal barrier	Require intermediate FLAT unless two strong opposite daily closes	Designed to minimize UP <-> DOWN whipsaw under event-driven or noisy reversals.
Shock and volatility vetoes	Gap >= 2 ATR, VOLX >= 1.75 with weak efficiency	Treats distorted environments as unreliable classification windows unless stronger persistence later appears.

20. Transition Truth-Table Appendix

Prev label	Candidate	Key condition	Immediate outcome	Needs confirm?	Comment
FLAT	UP	aggregate>=+40 and weekly>=+25	UP	No	Fast-track upgrade
FLAT	UP	ordinary UP candidate	Hold pending	Yes: 2 daily cycles	Standard upgrade
FLAT	DOWN	aggregate<=-40 and weekly<=-25	DOWN	No	Fast-track downgrade
FLAT	DOWN	ordinary DOWN candidate	Hold pending	Yes: 2 daily cycles	Standard downgrade
UP	FLAT	abs(aggregate)<15 or any_veto_flag	FLAT	No	Immediate degradation
DOWN	FLAT	abs(aggregate)<15 or any_veto_flag	FLAT	No	Immediate degradation
UP	DOWN	one opposite day only	FLAT intermediate	Yes	Direct reversal blocked
DOWN	UP	one opposite day only	FLAT intermediate	Yes	Direct reversal blocked
UP	DOWN	2 consecutive daily evals with aggregate<=-50 and weekly,daily<=-25	DOWN	No	Hard reversal allowed
DOWN	UP	2 consecutive daily evals with aggregate>=+50 and weekly,daily>=+25	UP	No	Hard reversal allowed
UP	UP	candidate matches previous	UP	No	Stable continuation
DOWN	DOWN	candidate matches previous	DOWN	No	Stable continuation

21. Final Locked Specification

Implement the Trend Identifier as a hierarchical, score-gated, persistence-filtered multi-timeframe regime classifier with the exact defaults and exact rule logic in Sections 3 through 20. The public output is exactly one of UP, FLAT, or DOWN. The internal state includes CLASSIFIABLE and UNCLASSIFIABLE, but the public payload is always emitted under the reference policy in Section 11.

Trend direction is determined from normalized return, baseline slope, baseline separation, breakout persistence, and swing structure. Path quality is determined from efficiency, acceptance, and overlap.

Noise is determined from gap shock, volatility expansion, anomaly, and liquidity penalties. Final labels are produced through explicit thresholds, explicit parenthesized conflict logic, explicit vetoes, explicit hysteresis, strict function contracts, deterministic missing-value handling, formal schema constraints, canonical reason-code enums, execution/runtime guarantees, and transition truth tables. Validation requirements and exact validation metrics defined in Section 17 are also part of the normative reference design.